

Homework 1:-

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1.D1:-

$$F = 1200 \frac{\text{Kmol}}{\text{hr}}$$

$$z_E = 0.3 \rightarrow 360 \frac{\text{Kmol E}}{\text{hr}}$$

$$z_W = 0.7 \rightarrow 840 \frac{\text{Kmol W}}{\text{hr}}$$

$$E: 0.3 \times 46.07 = 13.821 \frac{\text{Kg}}{\text{Kmol}}$$

$$W: 0.7 \times 18.01 = 12.607 \frac{\text{Kg}}{\text{Kmol}}$$

$$\text{Total: } 26.428 \frac{\text{Kg}}{\text{Kmol}}$$

$$\text{weight fraction of ethanol} = \frac{\text{weight of ethanol}}{\text{Total weight}} = \frac{13.821 \frac{\text{Kg}}{\text{Kmol}}}{26.428 \frac{\text{Kg}}{\text{Kmol}}} = 0.5229 = 52.3 \text{ wt\%}$$

$$\text{Flow rate: } 1200 \frac{\text{Kmol}}{\text{hr}} \times 26.428 \frac{\text{Kg}}{\text{Kmol}} = 31713.6 \frac{\text{Kg}}{\text{hr}} \times \frac{2.2 \text{ lb}}{\text{Kg}} = 69769.92 \frac{\text{lb}}{\text{hr}}$$

1.D3:-

$$F = 1200 \frac{\text{Kmol}}{\text{hr}}$$

$$0.3 E$$

$$0.7 W$$

$$E: \frac{30 \text{ Kg}}{46.07 \frac{\text{Kg}}{\text{Kmol}}} = 0.6512 \text{ Kmol}$$

$$W: \frac{70 \text{ Kg}}{18.01 \frac{\text{Kg}}{\text{Kmol}}} = 3.887 \text{ Kmol}$$

$$\text{Total moles} = 4.5379 \text{ Kmol}$$

$$\text{mol fraction of Ethanol: } \frac{0.6512 \text{ Kmol}}{4.5379 \text{ Kmol}} = 0.1435 = 14.35 \text{ mol\%}$$

$$\text{molar mass of the mixture} = \frac{100 \text{ Kg}}{4.5379 \text{ Kmol}} = 22.04 \frac{\text{Kg}}{\text{Kmol}}$$

$$\text{Total flow rate: } 1200 \text{ Kmol} \times 22.04 \text{ Kg} = 26448 \text{ Kg}$$

molar mass:- Ethanol: 46.07 Kg/Kmol

water: 18.01 Kg/Kmol

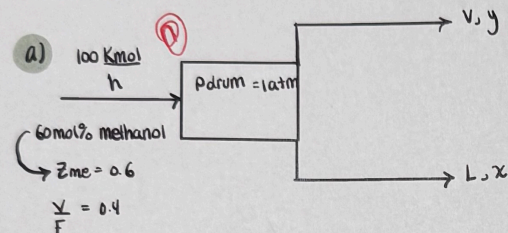
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basis: 100 Kg mixture. $\rightarrow 30 \text{ Kg E}$
 $\rightarrow 70 \text{ Kg W}$

molar mass:- Ethanol: 46.07 Kg/Kmol

water: 18.01 Kg/Kmol

3. (4pts) We are separating a mixture of methanol and water in a flash drum at 1.0 atm pressure.
- Feed is 60.0 mol% methanol, and 40.0% of the feed is vaporized. After setting up an operating equation, find the vapor and liquid mole fractions and flow rates. Feed rate is 100 kmol/h.
 - Repeat part (a) for a feed rate of 1500.0 kmol/h.
 - If the feed is 30.0 mol% methanol and we desire a liquid product that is 20.0 mol% methanol, what V/F must be used? For a feed rate of 1000.0 lbmol/h, find product flow rates and compositions.
 - We are operating the flash drum so that the liquid mole fraction is 45.0 mol% methanol. $L = 1500.0$ kmol/h, and $V/F = 0.2$. What must the flow rate and composition of the feed be?



Total M.B: $F = V + L$

Component M.B: $Fz = Vy + Lx$

Divide by F: $\left\{ y = \frac{Fz}{V} - \frac{Lx}{V} \right\}$ operating equation.

in terms of f using: $F = V + L \rightarrow L = F - V$

$\frac{L}{V} = \frac{F - V}{V} = \frac{1 - \frac{V}{F}}{\frac{V}{F}} = \frac{1 - f}{f}$

$f = \frac{V}{F}$

operating equation: $y = \frac{z}{f} - \left(\frac{1 - f}{f} \right) x$

$y = \frac{0.6}{0.4} - \left(\frac{1 - 0.4}{0.4} \right) x$

$y = 1.5 - 1.5x$ operating line.

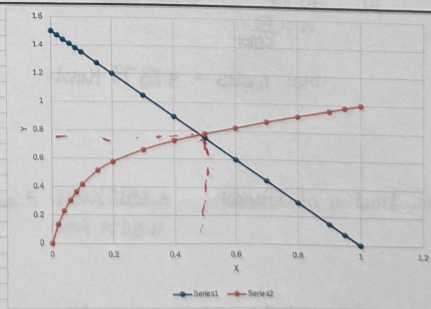
TABLE 2-8. Vapor-liquid equilibrium data for methanol-water ($p = 1$ atm) (mol %)

Methanol Liquid	Methanol Vapor	Temp., °C
0	0	100
2.0	13.4	96.4
4.0	23.0	93.5
6.0	30.4	91.2
8.0	36.5	89.3
10.0	41.8	87.7
15.0	51.7	84.4
20.0	57.9	81.7
30.0	66.5	78.0
40.0	72.9	75.3
50.0	77.9	73.1

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60.0	82.5	71.2
70.0	87.0	69.3
80.0	91.5	67.6
90.0	95.8	66.0
95.0	97.9	65.0

Operating line		Equilibrium curve	
x	y	x	y
0	1.5	0	0
0.02	1.47	0.02	0.134
0.04	1.44	0.04	0.23
0.06	1.41	0.06	0.304
0.08	1.38	0.08	0.365
0.1	1.35	0.1	0.418
0.15	1.275	0.15	0.517
0.2	1.2	0.2	0.579
0.3	1.05	0.3	0.665
0.4	0.9	0.4	0.729
0.5	0.75	0.5	0.779
0.6	0.6	0.6	0.825
0.7	0.45	0.7	0.87
0.8	0.3	0.8	0.915
0.9	0.15	0.9	0.958
0.95	0.075	0.95	0.979
1	0	1	1



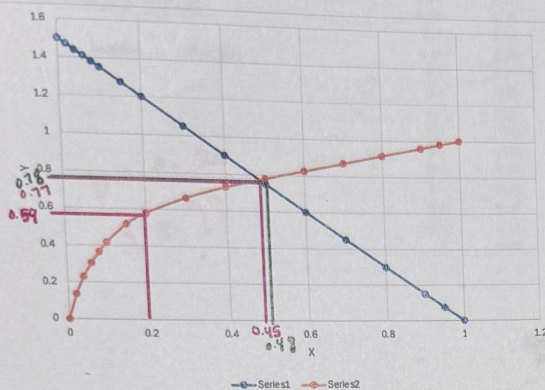
$y = ?$
 $x = ?$

Flow rates: $V = 100 \frac{\text{Kmol}}{\text{hr}} \times 0.4 = 40 \frac{\text{Kmol}}{\text{hr}}$

$F = V + L$

$L = F - V = (100 - 40) \frac{\text{Kmol}}{\text{hr}} = 60 \frac{\text{Kmol}}{\text{hr}}$

$X =$



$X = 0.48$
 $Y = 0.78$

$Fz = Vy + Lx$

$100(0.6) = (40)(0.78) + (60)(0.48) \Rightarrow 60 = 60$

b) $F = 1500 \frac{\text{Kmol}}{\text{hr}}$, $z = 0.6$, $f = 0.4$

$V = F \times 0.4 = 600 \text{ Kmol/hr}$

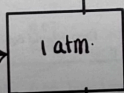
$F = V + L \rightarrow L = F - V = 900 \text{ Kmol/hr}$

$X = 0.48$, $Y = 0.78$

$V = F - L = 1500 - 900 = 600 \frac{\text{Kmol}}{\text{hr}}$

c) $F = 1000 \frac{\text{Kmol}}{\text{hr}} = 454 \frac{\text{Kmol}}{\text{hr}}$

$z = 0.3$



$L, x = 0.2$

$0.454 \times 0.2 = 90.9 \frac{\text{Kmol}}{\text{hr}}$

$y = \frac{z}{f} - \frac{(1-f)x}{f}$

From the graph: $y(x=0.2) = 0.59$

$y = \frac{z}{f} - \frac{(1-f)x}{f}$

$yf = z - x + xf$

$yf - xf = z - x$

$f = \frac{z-x}{y-x} = \frac{0.6-0.2}{0.59-0.2} = 0.256$

d) $F = ?$

1 atm

$f = \frac{y}{F} = 0.2$

$L = 1500 \frac{\text{kmol}}{\text{hr}}$

$x = 0.45$

$F = V + L$

$1 = \frac{y}{F} + \frac{L}{F} \rightarrow \frac{L}{F} = 1 - 0.2 = 0.8$

$\rightarrow F = \frac{L}{0.8} = \frac{1500}{0.8} = 1875 \frac{\text{kmol}}{\text{hr}}$

$V = F - L = 1875 - 1500 = 375 \frac{\text{kmol}}{\text{hr}}$

$Fz = Vy + Lx$

$z = \frac{y}{F} + \frac{L}{F} x$

$\quad \quad \quad y(x=0.45) = 0.77$

$z = (0.2)(0.77) + (0.8)(0.45) = 0.514$

4. (4pts) A methanol-water mixture is to be flash distilled at 1 atm. If the feed is 25 mole % methanol, what are the liquid and vapor compositions if:
- All of the feed is vaporized.
 - None of the feed is vaporized.
 - 1/3 of the feed is vaporized.
 - 2/3 of the feed is vaporized.

Solve all problems (a-d) graphically using the information in Table 2-8

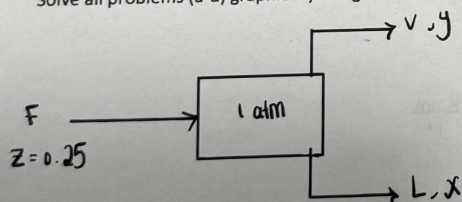


TABLE 2-8. Vapor-liquid equilibrium data for methanol water ($p = 1$ atm) (mol %)

Methanol Liquid	Methanol Vapor	Temp., °C
0	0	100
2.0	13.4	96.4
4.0	23.0	93.5
6.0	30.4	91.2
8.0	36.5	89.3
10.0	41.8	87.7
15.0	51.7	84.4
20.0	57.9	81.7
30.0	66.5	78.0
40.0	72.9	75.3
50.0	77.9	73.1

60.0	82.5	71.2
70.0	87.0	69.3
80.0	91.5	67.6

a) $\frac{y}{F} = 1 \rightarrow F = y$

$FZ = Vy + 0$

$Z = y = 0.25$

$L = 0, x = 0$

$y = 0.25$
 $x = ??$

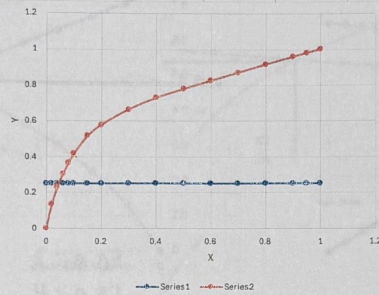
Q4-a

Operating line

x	y
0	0.25
0.02	0.25
0.04	0.25
0.06	0.25
0.08	0.25
0.1	0.25
0.15	0.25
0.2	0.25
0.3	0.25
0.4	0.25
0.5	0.25
0.6	0.25
0.7	0.25
0.8	0.25
0.9	0.25
0.95	0.25
1	0.25

Equilibrium curve

x	y
0	0
0.02	0.134
0.04	0.23
0.06	0.304
0.08	0.365
0.1	0.418
0.15	0.517
0.2	0.579
0.3	0.665
0.4	0.729
0.5	0.779
0.6	0.825
0.7	0.87
0.8	0.915
0.9	0.958
0.95	0.979
1	1



write answers please.

b) $\frac{y}{F} = 0$

$v = 0, y = 0, x = 0.25$

$x = 0.25$
 $y = ??$

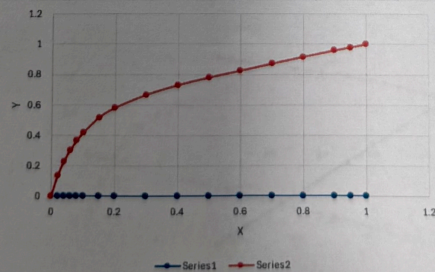
Q4-b

Operating line

x	y
0	0
0.02	0
0.04	0
0.06	0
0.08	0
0.1	0
0.15	0
0.2	0
0.3	0
0.4	0
0.5	0
0.6	0
0.7	0
0.8	0
0.9	0
0.95	0
1	0

Equilibrium curve

x	y
0	0
0.02	0.134
0.04	0.23
0.06	0.304
0.08	0.365
0.1	0.418
0.15	0.517
0.2	0.579
0.3	0.665
0.4	0.729
0.5	0.779
0.6	0.825
0.7	0.87
0.8	0.915
0.9	0.958
0.95	0.979
1	1

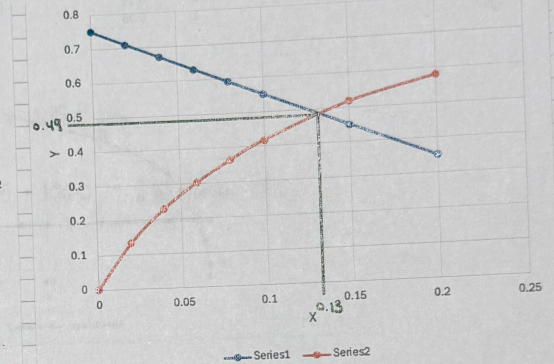
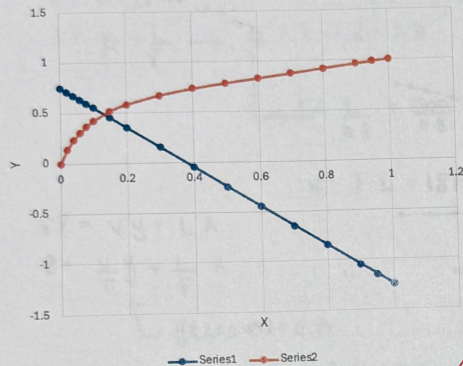


c) $\frac{K}{f} = \frac{1}{3} \rightarrow$

$$y = \frac{z}{f} - \frac{(1-f)x}{f}$$

$$y = \frac{0.25}{\frac{1}{3}} - \frac{(1-\frac{1}{3})x}{\frac{1}{3}}$$

$$y = 0.75 - 2x$$



$$x = 0.13$$

$$y = 0.49$$

$$z = \frac{K}{f}y + \frac{1}{f}x$$

$$z = \frac{1}{3}(0.49) + (1 - \frac{1}{3})0.13 = 0.25$$

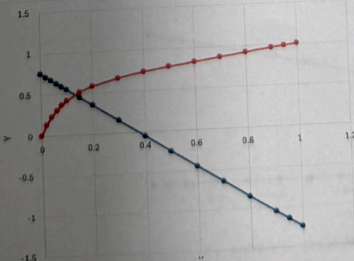
Q4-c

Operating line

x	y
0	0.75
0.02	0.71
0.04	0.67
0.06	0.63
0.08	0.59
0.1	0.55
0.15	0.45
0.2	0.35
0.3	0.15
0.4	-0.05
0.5	-0.25
0.6	-0.45
0.7	-0.65
0.8	-0.85
0.9	-1.05
0.95	-1.15
1	-1.25

Equilibrium curve

x	y
0	0
0.02	0.134
0.04	0.23
0.06	0.304
0.08	0.365
0.1	0.418
0.15	0.517
0.2	0.579
0.3	0.665
0.4	0.729
0.5	0.779
0.6	0.825
0.7	0.87
0.8	0.915
0.9	0.958
0.95	0.979
1	1

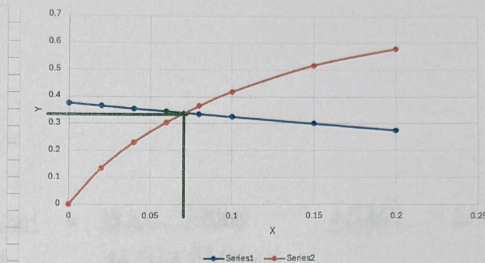
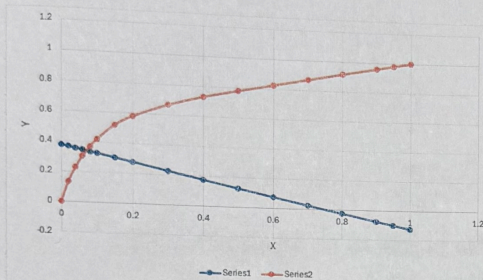


d) $\frac{y}{f} = \frac{2}{3}$

$$y = \frac{z}{f} - \frac{(1-f)x}{f}$$

$$y = \frac{0.25}{(2/3)} - \frac{(1-2/3)x}{(2/3)}$$

$$y = 0.375 - 0.5x$$



$x = 0.07$

$y = 0.32$

$$z = \frac{y}{f} + \frac{1-f}{f} x$$

$$= \left(\frac{2}{3}\right)(0.32) + \left(1 - \frac{2}{3}\right)(0.07) = 0.24$$

Q4-d

Operating line

x	y
0	0.375
0.02	0.365
0.04	0.355
0.06	0.345
0.08	0.335
0.1	0.325
0.15	0.3
0.2	0.275
0.3	0.225
0.4	0.175
0.5	0.125
0.6	0.075
0.7	0.025
0.8	-0.025
0.9	-0.075
0.95	-0.1
1	-0.125

Equilibrium curve

x	y
0	0
0.02	0.134
0.04	0.23
0.06	0.304
0.08	0.365
0.1	0.418
0.15	0.517
0.2	0.579
0.3	0.665
0.4	0.729
0.5	0.779
0.6	0.825
0.7	0.87
0.8	0.915
0.9	0.958
0.95	0.979
1	1

